

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460285.6635 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.056 +/- 0.053
Transit depth (Rp/Rs)^2	0.31 +/- 0.59 [%]
Orbital Inclination (inc)	89.53 +/- 1.73
Airmass coefficient 1 (a1)	1.2529 +/- 0.01
Airmass coefficient 2 (a2)	-0.18 +/- 1.74
Scatter in the residuals of the lightcurve fit is	0.79 %
Best Comparison Star	None
Optimal Aperture	3.4
Optimal Annulus	12.43
Transit Duration (day)	0.1121 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.056 $\pm$ 0.053 vs 0.1489 (deviation 1.28 σ)
Duration fit vs published	0.1121 d vs 0.1304 d (deviation 3.43 σ)
Mid-time O-C	-8.3 min
Depth SNR	0.8
Residual scatter	0.79 %

Light curve

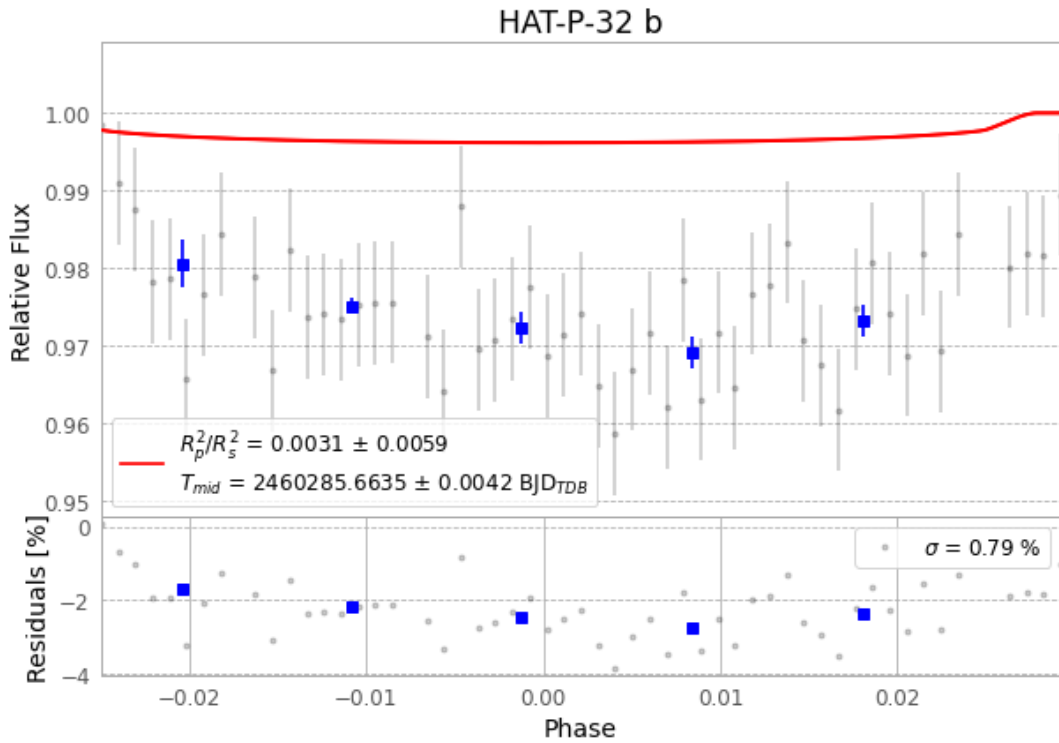


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-12-06.png (SHA-256 befda65a5b999242...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [437, 339], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ae7b8ee44b4a0915...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

53 FITS frames (35 MB), HATP-32231207023015.FITS → HATP-32231207051815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-231207.log (a13e5f996fca...), FinalLightCurve_HAT-P-32 b_2023-12-06.png (befda65a5b99...), FinalLightCurve_HAT-P-32 b_2023-12-06.csv (c0cd2b32844a...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 15:44Z.